MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS MARK 83 MODS AND MARK 85 MODS (LOADED ASSEMBLIES)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the motors described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Motors, Mark 83 Mod 0 and Mark 85 Mod 0, hereinafter referred to as the "rocket motors" (see 6.1).

1.2 Classification. The rocket motors shall be of the following types as specified in the contract or purchase order:

Type I	Mark 83 Mods (low Yaw)
Type II	Mark 85 Mods (high Yaw)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-10464	Cans, Hermetic Sealing, Metal Light Gage, Tear-Strip Type

Naval Air Systems Command (Code Ident 30003)

AS 3885	Propellant, N-47 (U)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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DRAWINGS

Naval Air Systems Command (Code Ident 30003)

946AS100 Rocket Motor Assembly, Mark 83 Mod 0

989AS100 Rocket Motor Assembly, Mark 85 Mod 0

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 83 Mod 0 Rocket Motor shall conform to Drawing 946AS100 and the Mark 85 Mod 0 Rocket Motor shall conform to Drawing 989AS100.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the requirements of table I.

TABLE I. Propellant N-47 (AS 3885).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	149	-	149	-	149	-
E_m (elongation at S_m) %	77	21.3	-	10	-	18.5	-
Y_m (Young's modulus) psi	77	960	1800	960	2862	960	2880
B_r strand burn rate at 1000 psi) in/sec	77	0.69	0.84	0.69	0.89	0.69	0.89

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3885 when tested in accordance with MIL-A-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to table II.

TABLE II. Ballistic performance requirements.

Parameter	Mk 83 Mods (low yaw thruster)				Mk 85 Mods (high yaw thruster)			
	−40° ± 5°F		160° ± 5°F		−40° ± 5°F		160° ± 5°F	
	Min	Max	Min	Max	Min	Max	Min	Max
Rocket motor impulse (lb-s)	10.3[1]	12.4	10.3	12.4	21.0[1]	24.1	21.0	24.1
Action time (ms)	-	160[1]	79	-	-	174[1]	81	-
Burn time (ms)	-	147	70[1]	-	-	161	72[1]	-
Maximum rocket motor thrust (lb)	80[1]	115	100	160[1]	120[1]	235	160	315[1]
Time to 70% maximum rocket thrust (ms)	-	30	-	30	-	30	-	30
Rocket motor ignition time (ms)	-	20	-	20	-	20	-	20

[1] Escape system requirements.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Examinations and tests.

4.1.1 Preproduction/production.

4.1.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: −40° ± 5°F
b. Hot: 160° ± 5°F

4.1.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 3 hours
b. Temperature cycling: not less than 3 hours

5. PACKAGING

5.1 Preservation-packaging.

5.1.1 Packaging. Four rocket motors shall be placed in a hermetically sealed can in accordance with MIL-C-10464, Type I. The package shall be adequately cushioned to protect the rocket motors against handling and shipping shocks and vibration and shall comply with the applicable requirements of MIL-P-116, method 1A-5 and 49 CFR 171-190.

5.2 Marking. In addition to those required by the general specification, marking shall include "Explosive Power Device, Class C".

6. NOTES

6.1 Intended use. The Rocket Motors Mk 83 Mod 0 and Mark 85 Mod 0 are intended for use as a means of insuring proper trajectory divergence of the aircrew during an escape for the S-3 aircraft escape system. The Rocket Motor Mk 83 Mod 0 is also used in the T-A4 aircraft.

6.2 Definition.

6.2.1 Rocket motor lot. A rocket motor lot is all those motors cast from one batch of propellant.

6.3 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/6A(AS)	AIRCREW ESCAPE PROPULSION SYSTEMS; Rocket Motors MARK 88 MODS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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